

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Explain with block diagram AV micro-controller. (CO-2)

Q.24 Explain the interfacing of relay and sensors.(CO-3)

Q.25 Write short note on the following.

(a) Troubleshoot of microcontrollers. (CO-4)

(b) Real-time operating system. (CO-1)

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Roll No.

4th Semester/ Artificial Intelligence & Machine Learning Subject : Embedded System and Design

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Which factor affects the reliability of embedded systems? (CO-1)

- a) Temperature b) Humidity
- c) Power supply d) All of the above

Q.2 Which AVR microcontroller instruction is used to load a value into a register? (CO-2)

- a) LDI b) LD
- c) MOV d) LOAD

Q.3 Troubleshoot meaning in microcontrollers is: (CO-4)

- a) To find and fix issues
- b) To create troubles
- c) To interface with appliances
- d) To write a program

- Q.4 What is the selection criteria for a micro processor? (CO-1)
- a) Processing power b) Memory size
c) I/O capabilities d) All of the above
- Q.5 Which pin of a microcontroller is used to connect an LED? (CO-3)
- a) VCC b) GND
c) Digital output pin d) Analog input pin
- Q.6 Which applications uses embedded systems?(CO-4)
- a) Desktop computers
b) Mobile phones
c) Industrial control systems
d) All of the above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

- Q.7 The PIC microcontroller instruction used to jump to a label is _____. (CO-2)
- Q.8 The display commonly used to display numerical data is _____. (CO-3)
- Q.9 The brain of a microcontroller is the _____. (CO-1)

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- Q.10 The AVR microcontroller instruction used to set a bit in register is _____. (CO-2)
- Q.11 CBI instruction in AVR microcontroller is used to _____ a bit in a register. (CO-2)
- Q.12 VCC pin of microcontroller is connected to _____. (CO-1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

- Q.13 Write the characteristics of embedded systems. (CO-1)
- Q.14 Explain the buzzer interfacing with microcontroller. (CO-3)
- Q.15 Explain the selection criteria of microcontroller. (CO-1)
- Q.16 What is the significance of interfacing? (CO-3)
- Q.17 Compare PIC and AVR microcontrollers. (CO-2)
- Q.18 How can we repair microcontroller based appliances? (CO-4)
- Q.19 Explain the embedded system architecture. (CO-1)
- Q.20 Explain the factors affecting embedded systems. (CO-1)
- Q.21 Write the uses of embedded systems. (CO-2)
- Q.22 Write a short note on PIC architecture. (CO-2)

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